DNA Technology and Bioinformatics

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

DNA Technology and Bioinformatics (A13863)

Short Title:	DNA Technol. & Bioinformatics		
Faculty	Science and Computing		
Department	Science		
Cluster	Biology		
Author	ROMAHONY		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module will focus on the techniques associated with molecular biology/biotechnology and the use of bioinformatics tools and databases.

Programmes

MOLB-0004	BSc (Hons) in Applied Biology with Quality Management (WD_SQMBI_B)	4 / 1 / M
MOLB-0004	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	4 / 7 / M

Indicative Content

- Principles of DNA cloning and techniques
- Construction and screening of cDNA and genomic libraries
- Gene and genome sequencing; applications and next-generation sequencing technologies
- Gene expression and regulation and the role of epigenetic modifications; DNA tools relevant to epigenomics
- Advanced DNA techniques/tools in the study of gene detection/ expression/ genotyping/ polymorphisms/DNA profiling, such as DNA microarrays, SNP arrays and real-time PCR
- Emerging DNA technologies such as genome editing and digital PCR
- Bioinformatics: basic sequence analysis; DNA and protein databases; Clustal/BLAST-based searching and alignments; cladograms and phylogenetic trees; protein sequence/function determination

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Comment on the principles of DNA cloning and evaluate various vector and library systems.
2. Evaluate the techniques used to sequence individual genes and entire genomes.
3. Compare the methods used to study individual genes and entire genomes, such as detection, expression and function.
4. Distinguish emerging methodologies relevant to molecular biology.
5. Manage, interpret and evaluate various bioinformatics analysis tools and various DNA technology practical techniques.

Learning and Teaching Methods

- Lectures using a blended learning approach.
- Lab-based practicals.
- Computer-based bioinformatic classes.
- A learning and teaching strategy appropriate to level 8 will be adopted throughout with opportunities for formative assessment to encourage reflective, critical thinking.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4
Continuous Assessment	50%	
Practical	50%	5

Assessment Criteria

<40%: No understanding of basic module content. Unable to carry out critical thinking and analysis.
40%-49%: Able to understand basic module concepts. Has difficulty with critical thinking and problem solving.
50%-59%: A good understanding of module content with demonstration of some critical thinking and problem solving skills.
60%-69%: A good understanding of module content and good critical thinking and problem solving skills.
70%-100%: Excellent in all areas.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- "Moodle used to provide links to useful websites." https://moodle.wit.ie/login/index.php
- Brown, T.A. _Gene Cloning and DNA analysis: An introduction_. 7th ed. UK: Wiley, 2016.

Supplementary Material

- Hartl, D. and M. Ruvolo. _Genetics: Analysis of Genes and Genomes_. MA USA: Jones and Barlett, 2011.
- Lesk, A. _Introduction to Bioinformatics_. 4th ed. UK: Oxford, 2008.

Requested Resources

- Room Type: Computer Lab
- Room Type: Biology Lab
- Lecture Room: Loose Seated